SCH: INT: Collaborative Research: Prediction of Preterm Birth in Nulliparous Women

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Summary Prediction of Preterm Birth (PTB) has been an exceedingly challenging problem, predominantly due to the inherent complexity of its multifactorial etiology and the lack of approaches capable of integrating and interpreting large multidisciplinary data. It is a major long-lasting public health problem with heavy emotional and financial consequences to families and society [22, 64]. PTB is the leading cause of mortality and long-term disabilities among neonates. Most studies to date have examined individual risk factors through univariate analyses of their coincidence with PTB.

Our previous work [NSF Eager 1454855, 1454814] developed predictive models for PTB based on non-genetic maternal attributes [103, 101]. A particularly challenging population to determine PTB risk is first time mothers (nulliparous women) due to the lack of prior pregnancy history. An important question is to know whether factors other than history of PTB can be used to identify a nullipara patient at risk. Recently published work has revealed the contribution of common variants to gestation length and has identified six gene regions associated with PTB [110]. We plan on devising longitudinal risk prediction methods for PTB that use every piece of available data. We will address three important gaps in current literature as our three project objectives: study nulliparous women and their risk for PTB; combine genetic factors with other clinical factors to determine risk; and use longitudinal data and models to optimize scheduling of patient visits, testing and treatment. We will focus on a recently released NIH-NICHD dataset called nuMoM2b, which is a prospective cohort study of a racially/ethnically/geographically diverse population of 10,038 nulliparous women with singleton gestation.

Intellectual Merit Our objectives are as follows: (1) **Longitudinal Preterm Birth Prediction:** We will first build a series of accurate prediction models for PTB using the nuMoM2b dataset. Such models will handle the challenges common to medical datasets including (a) imbalance in the classes, (b) missing data, and (c) disparity in data collection. We will achieve this by designing an objective function for Support Vector Machines that captures and corrects for these issues. Second, by leveraging the availability of patient future data, our Learning Under Privileged Information (LUPI)-based approach [66] will significantly increase the rate of convergence of the algorithms and improve prediction with less data. Our transformative approach is well-suited for medical datasets that are both limited by the number of patients and inherently include the challenges mentioned above. (2) **Combining clinical and genetic features for risk prediction**: In this aim we tackle questions of causality between the genetic information and its various forms of phenotypic implications by leveraging the phenotypically rich nuMoM2b dataset. We will first apply standard GWAS analysis to apply new insight regarding the changing patterns of genetic association as additional phenotypic data is accumulated as well as serve as a baseline. We will then seek to develop improved analysis of involvement of genetic contributions in PTB. (3) **Clinical and social impact**: We plan to assess the effectiveness of the methods in clinical practice by: (a) testing the effectiveness of the longitudinal models produced in objective 1 and 2 on existing clinical data at the New York Presbyterian Hospital. (b) building a sequential decision making model; this includes optimizing the scheduling of patient visits and diagnostic testing tailored for different classes of patients.

Broader Impact Over 26 billion dollars are spent annually on the delivery and care of the 12% of infants who are born preterm in the United States. A crucial challenge is to identify women who are at the highest risk for very early preterm birth and to develop interventions. Equally important, would be the ability to identify women at the lowest risk to avoid unnecessary and costly interventions. Our project has the potential to advance knowledge about this long-lasting public health problem. We will recruit female and minority students to this project. Research results will be disseminated through courses, conference and journal papers, and through undergraduate research and graduate students' theses.

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1 Introduction

Premature or preterm birth (PTB) is defined as birth before 37 weeks of gestation. It is a major long-lasting public health problem with heavy emotional and financial consequences to families and society [64, 22]. PTB is the leading cause of mortality and long-term disabilities among neonates. These range from visual/hearing impairment, cerebral palsy and mental retardation to increased likelihood of cardiovascular disease, hypertension, and diabetes later in life [109]. Furthermore, over 26 billion dollars are spent annually on the delivery and care of the approximately 12% of infants who are born preterm in the United States [11]. A crucial challenge is to identify women who are at the highest risk for extremely preterm birth in order to facilitate interventions. Equally important, would be the ability to identify women at the lowest risk to avoid unnecessary and costly interventions.

Figure 1: Picture of a 23 weeks preemie

Risk factors Approximately 30% of preterm deliveries are indicated based on maternal or fetal indications. The remaining 70%, spontaneous PTB (SPTB) occur following the onset of spontaneous preterm labor, prelabor Premature Rupture Of the Membranes (pPROM) or cervical insufficiency [36]. Spontaneous preterm labor is a heterogeneous condition, the final common product of numerous biologic pathways that include immune, inflammatory, neuroendocrine, and vascular processes [11]. Those judged at risk for PTB are typically treated by prenatal administration of progesterone 17 OHPC (IM progesterone) [4, 34]. Epidemiological investigations have largely associated single factors with PTB. Of the many risk factors for preterm labor, a prior history of preterm delivery is the most predictive with a recurrence risk as high as 50% depending on the number and gestational age of previous deliveries [36, 61]. In the United States, women classified as black, African-American, and Afro-Caribbean are at an increased risk [36]. Other risk factors include low socioeconomic status, extremes in age, single marital status [91, 93]; low pre-pregnancy body mass index, [43]; and high-risk behaviors during pregnancy (e.g., tobacco, cocaine and heroin use). Psychological factors that have been proposed to increase the risk of PTB include depression [41], and high levels of stress during pregnancy [23, 111]. Obstetrical conditions associated with the onset of spontaneous preterm labor are heterogeneous and include intrauterine infection, uteroplacental ischemia and hemorrhage, vascular lesions of the placenta, uterine over distention and cervical insufficiency [37, 39, 78, 94]. Additional pregnancy factors associated with preterm delivery include closely spaced gestations [21], and multiple gestations [36]. Assisted reproductive technologies may also constitute a risk factor, for unknown reasons [10, 68]. Finally, environmental toxins have been suggested as risk factors, though results are conflicting [11]. Exposure to tobacco smoke appears to carry the greatest risk [50, 48]; other pollutants studied include exposure to sulfur dioxide and particulates [58, 77]. Besides these, there is some evidence for a genetic contribution: women with sisters who gave birth preterm had an 80% higher risk of delivering preterm [107] and grandparents of women delivering prematurely were more likely to have been born preterm themselves [69]. Recent published work revealed contribution of common variants to gestation length and has identified six genes regions associated with PTB [110].

There are other critical factors that have been used for predicting PTB by some practices. It has been shown [35] that the odds ratio of SPTB was highest for fetal fibronectin followed by short cervix and history of prior PTB. In fact, these factors along with bacterial vaginosis were more strongly associated with early than late SPTB. Systematic studies of multiple published studies [44, 28] showed that cervical length (threshold of 25 mm) measured by transvaginal ultrasonography in asymptomatic (no uterine tightening/contractions) high-risk women is highly predictive of spontaneous preterm birth before 35 weeks. However, in practice, prior history of preterm delivery is used as the most predictive indicator of PTB in most clinical settings.

Systems for predicting preterm birth Papiernik proposed in the late sixties an empirical method for estimating the risk of premature delivery denoted "The Coefficient of Premature Delivery Risk (CPDR)" [65]. In this approach, maternal characteristics are grouped into four series of comparable variables (social status, obstetric history, work conditions, pregnancy characteristics) in a two-dimensional table. A number of points varying from 1 to 5 according to the degree of their importance is assigned to each characteristic. The sum of the points gives the risk of premature delivery. Papiernik's risk table was later modified and used in the risk of preterm delivery (RPD) system proposed in [29] (Figure 2). Further assessment of the prediction performance of Creasy's table [32] on another population has shown low performance.

A graded risk system was proposed in [60] in the context of the NICHD MFMU preterm prediction study. The results of a multivariate logistic regression were modest: sensitivity of 24.2% and 18.2%; Specificity of 28.6% and 33.3%, respectively for nulliparous (mothers-to-be) and multiparous women.

This study was a first step toward combining factors showing promise that using more sophisticated techniques could better identify women at-risk of PTB. Today, there is no widely tested risk scoring/prediction system that combines PTB factors [30]. The NIH [109] and particularly the NICHD [62, 63] have recognized the need to discover the multifactorial etiologies of PTB and to use these to accurately predict at-risk pregnancies.

Figure 2: Creasy's RPD table [29]

Points	Socioeconomic status	Past history	Daily habits	Current pregnancy
1	2 children at home Low socioeconomic status	1 abortion Less than 1 year since last birth	Work outside home	Unusual fatigue
2	Younger than 20 years Older than 40 years Single parent	2 abortions	More than 10 cigarettes per day	Less than 13 kg gain by 32 weeks' gestation Albuminuria Hypertension Bacteriuria
3	Very low socioeconomic status Shorter than 150 cm Lighter than 45 kg	3 abortions	Heavy work Long tiring trip	Breech at 32 weeks Weight loss of 2 kg Head engaged Febrile illness
4	Younger than 18 years	Pyelonephritis		Metrorrhagia after 12 weeks' gestation Effacement Dilatation Uterine irritability
5		Uterine anomaly Second-trimester abortion DES exposure		Placenta previa Hydramnios
10		Premature delivery Repeated second-trimester abortion		Twins Abdominal surgery

* Score is computed by addition of the number of points given any item. 0–5 = low risk; 6–9 = medium risk; ≥ 10 = high risk.

Goodwin et al. have explored the use of data mining techniques to predict preterm labor. In early work [108], they show that it is feasible to generate expert systems (knowledge-base) rules for prediction of preterm delivery. In subsequent work [38], data from the Duke University medical center in North Carolina consisting of 19,970 patient records and 1,622 variables was studied using data mining techniques. In this paper, the authors claim to have identified a parsimonious set of seven demographic variables (maternal age, marital status, race, education, patient insurance category, county and religion) that play a more critical role as predictors of preterm birth for their data set. While these results are interesting, there are concerns whether the sampling of a particular demographic (academic medial center) would be representative of more general population data. Furthermore, their experiment procedure is unclear – for example, the AUC could have been obtained on a validation set or an unseen test set; consequently it is difficult to reproduce their results. Courtney et al. [26] describe a secondary analysis showing that the demographic preterm prediction model generated by [38] generalizes to a broader population but with a modest accuracy.

Our preliminary work developed predictive scores for PTB based on non-genetic maternal attributes [20, 103, 101]. A particularly challenging population to determine PTB risk is first time mothers (nulliparous women) due to the lack of prior pregnancy history. For all women who have had a prior preterm birth, the current standard of care is to use 17 OHPC (IM progesterone) [5]. Hence, first-time mothers representing 40% of the population of pregnant women, are simply not treated. A reliable prediction of PTB for those mothers may substantially reduce the incidence of their PTB and its consequences - perhaps by as many as the current 200,000 per year in the US alone. An important question is to know whether factors other than history of PTB can be used to identify a nullipara patient at risk and what is the individual contribution of each of the risk factors.

For the first time in PTB research, we will develop a dynamic risk prediction system for preterm birth that will address these important two gaps; that is study nulliparous women and their risk to PTB and include genetic factors that have been lacking in most studies and never combined with other risk factors. We will focus on a new dataset, the nuMoM2b (Nulliparous Pregnancy Outcomes Study Monitoring Mothers-to-be), focuses on pregnancy outcomes for 10,038 nulliparous mothers with singleton gestations just made available by the NIH-NICHD specifically for that challenging class of PTB [42, 3].

2 Proposal at a glance

2.1 A Compelling Application from a Machine Learning Perspective

With support from the National Science Foundation [NSF Eager 1454855, 1454814], our team conducted a preliminary study on the prediction of preterm birth from clinical, non-genetic patient attributes. We demonstrated through a secondary analysis of the NICHD MFMU Preterm Prediction Study dataset [59, 20],(Which we call the MFMU dataset) significant improvement in predicting preterm birth compared to previous prediction performance on the same data and developed models that integrate heterogeneous risk factors. Previous results on this dataset using a multivariate logistic regression model show a sensitivity of 24.2% and 18.2%, and specificity of 28.6% and 33.3%, for nulliparous and multiparous women respectively. The improvement is of 30% in specificity and sensitivity superior to the state-of-the-art.

The focus of this study will be on the nuMoM2b (Nulliparous Pregnancy Outcomes Study Monitoring Mothers-to-be) study and data collection that was initiated in 2010 by the NIH, NICHD [3].

In terms of computational methodology, PTB poses analytic challenges that motivate novel approaches. First, the sparsity of cases (about 12% of mothers) limit methods to allow unbalanced datasets with a small number of positive examples.This phenomenon is accentuated considering the different categories of PTB by gestational age at birth: extreme (<28 weeks), severe (28-31 weeks), moderate (32-33 weeks) and late (34-36 weeks) [36]. We also need to consider spontaneous PTB and indicated PTB cases separately.

Second, pregnancy is a dynamic and fast-changing process; different gestational ages at birth are believed to be driven by different etiologies, which makes PTB prediction temporal. That is, one single model will not fit all cases – one needs longitudinal prediction models.

Third, the attributes of the pregnancy involve time-dependent information, that becomes gradually available for each example/subject. This kind of situation was termed privileged information in [66] and [99] where an extension to the existing machine learning paradigm that incorporates specific elements of teaching was presented. This advanced learning paradigm, called "Learning Using Privileged Information" (LUPI), assumes that the teacher supplies the student (algorithm) with additional information about each training example. However, this information is not available for any of the test examples (hence it is privileged). The goal of LUPI is to increase the rate of convergence of the algorithm [66].

The goal of this proposal is to pursue our work and devise robust longitudinal risk prediction methods for PTB that use every piece of available data. We will address three important gaps in current literature: study nulliparous women and their risk for PTB; combine genetic factors with other clinical risk factors;

and use longitudinal data and models to optimize scheduling of patient visits, testing and treatment. These form the three aims of the project.

We anticipate our project will result in accurate and clinically useful predictive models that ultimately will translate into an improved diagnostic and care of nulliparous women with a long-term health outcome for their babies.

2.2 Intellectual Merits

Our aims are as follows:

(1) **Longitudinal Preterm Birth Prediction:** We will first build a series of accurate prediction models for PTB using the NuMom2b dataset. Such models will handle the challenges common to medical datasets including (a) imbalance in the classes, (b) missing data, and (c) data collection bias. We will achieve this by designing an objective function for a maximum margin classifier (Support Vector Machine) that captures and corrects for these issues. Second, by leveraging the availability of patient future data, our Learning Under Privileged Information (LUPI)-based approach will significantly increase the rate of convergence of the algorithms and improve prediction with less data. We not only build models for different time points using historical data, but also use the future data that is available in the training set to build a better classifier at a particular time point. Our transformative approach is well-suited for medical datasets that are both limited by the number of patients and inherently include the challenges mentioned above.

(2) **Combining clinical and genetic features for risk prediction**: In this aim we tackle questions of causality between the genetic information and its various forms of phenotypic implications by leveraging the phenotypically rich nuMoM2b dataset. We will first apply standard GWAS analysis to apply new insight regarding the changing patterns of genetic association as additional phenotypic data is accumulated as well as serve as a baseline. We will then seek to develop improved analysis of involvement of genetic contributions in PTB.

(3) **Clinical and social impact**: We plan to assess the effectiveness of the methods in clinical practice by: (a) testing the effectiveness of the longitudinal models produced in objective 1 and 2 on existing clinical data at the New York Presbyterian Hospital. (b) building a sequential decision making model; this includes optimizing the scheduling of patient visits and diagnostic testing tailored for different classes of patients.

2.3 Broader Impacts

The proposed research has the potential to make revolutionary progress in prediction and prevention of preterm birth. It will specifically benefit maternal-fetal medicine research to help identify women at risk of PTB. This has the potential to solve a long-lasting public health problem. Specifically, first-time mothers (nulliparous) representing 40% of the population of pregnant women, are simply not treated [109]. A reliable prediction of PTB for those mothers may substantially reduce the incidence of PTB and its consequences - perhaps by as many as 160,000/year in the US (with PTB rate=10%, #births in the US/year $\approx$ 4 million). This research would be readily applicable to other adverse pregnancy outcomes such as Intra Uterine Growth Restriction (IUGR) and Pre-eclampsia. Together with PTB, these outcomes account for most of neonatal mortality and morbidity. Furthermore, our research has the potential to generally impact other health-related issues where prediction will benefit prevention in clinical practice. The main education focus of our proposal is primarily through teaching, students mentoring and research training. The PI teaches Artificial Intelligence both on campus and online. Co-PI Pe'er is the Chair of the Health Analytics Institute within the Data Science Institute (DSI) which aims to research and education in learning from large amounts of data. PI Salleb-Aouissi is an affiliated member and together they have access to the Data Science Institute resources and faculty affiliates who work in medical informatics. The PIs will integrate material from this research into Artificial Intelligence (AI) courses both at Columbia University and Cooper Union. The PIs

have successfully trained students at all levels for several years. PI Salleb-Aouissi has offered a popular AI course on the edX platform which has attracted 250,000 students [1]. They plan to work with two graduate students and involve undergraduate students. We will recruit female and minority students to this project with high potential for social-impact. Research results will be disseminated through conference and journal papers, and through graduate students' theses. We will make our code publicly available.

3 Preterm birth data

We have access to two perinatal datasets:

a) **MFMU data.** The MFMU dataset which was the basis for our preliminary investigations was from an observational prospective study of 3,073 women with singleton pregnancies recruited at less than 24 weeks gestational age. Of the women enrolled, 2,929 participated in the study at the 10 participating MFMU centers across the United States between October 1992 and July 1994. There were 1,711 multiparous and 1,218 nulliparous women. The incidence of spontaneous preterm birth was 10.3% overall 8.2% for nulliparous and 11.9% for multiparous women (Mercer et al., 1996). Participating women in this study were followed up by research nurses during four visits at 24 (time T0), 26 (time T1), 28 (time T3) and 30 (time T4) weeks gestation for screening tests.

b) **nuMoM2b data: An innovative dataset**

In this cohort, patient demographic characteristics, clinical characteristics (e.g., blood pressure), sonographic findings (e.g., cervical length, uterine artery Doppler waveforms), exposures (e.g., diet, stress), and outcomes defined a priori are highly detailed and were collected by trained research personnel who were centrally certified in the data collection specifically required for this cohort.

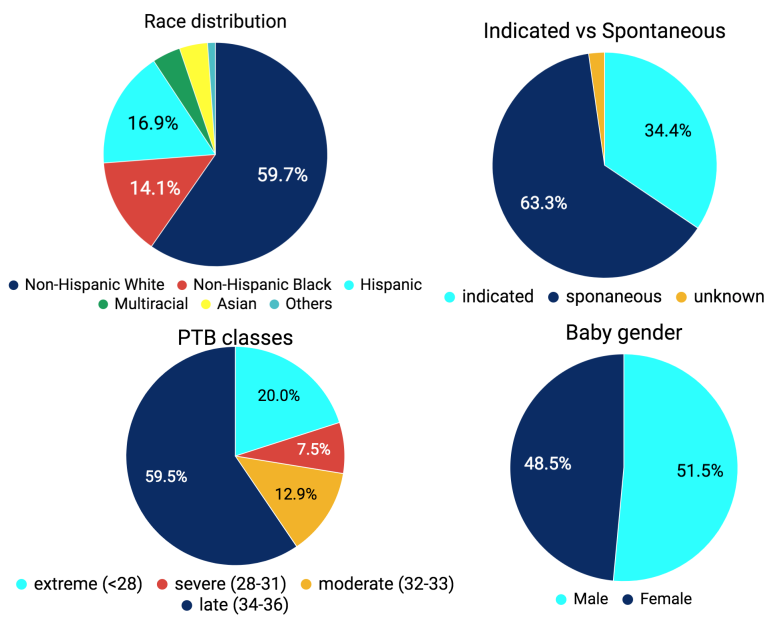


Figure 3: Preliminary statistics in the nuMoM2b data

Analytes from different sources (e.g., blood, cervico-vaginal secretions), moreover, were obtained at different time points in pregnancy according to standardized collection and processing protocols [42]. Thus, this cohort of nulliparous pregnant women is larger and better characterized across multiple domains than any prior population, allowing a more comprehensive and accurate approach to preterm birth prediction than previously has been possible.

Prior attempts to predict preterm birth and PE have been hindered by study populations that were small and/or poorly characterized both with respect to important clinical, psychosocial, and behavioral attributes and critically important elements of the birth outcome (e.g., spontaneous versus indicated preterm birth). Additionally, most studies have been limited by the timing and variety of biological samples available for analyses. In the proposed work we will use the nuMoM2b dataset which is from a meticulous, prospective cohort study of a racially/ethnically/geographically diverse population of 10,038 nulliparous women with singleton gestations. Dr. Wapner is a long-term PI on nuMoM2b projects and a member of the consortium. Figure 3 and 4

show initial statistics for mothers and babies.

During the nuMoM2b study, three data collection time points occurred during gestation corresponding with routine clinical visits: study visit 1 (6+0 through 13+6 weeks gestation), study visit 2 (16+0 through 21+6 weeks gestation), and study visit 3 (22+0 through 29+6 weeks gestation). The correlation with regular visits was chosen to facilitate ultimate translation of the findings into clinical care if predictive markers were to be established.

During study visit 1, blood pressure, height, weight, and circumferences of the waist, hip and neck were measured. Detailed interviews were performed to collect demographic characteristics, medical history, family history, physical activity levels, alcohol, tobacco, and drug use, and nutritional intake. In addition, multiple psychosocial assessments were performed, including with regard to perceived stress, reactions to racism, depression, sleep, anxiety, perceived social support, and difficulties in pregnancy. Sonographic CRL also was recorded for all participants. During study visit 2, measurements of blood pressure and weight were repeated and updates on physical activity level, and exposure to alcohol, drugs, and tobacco were obtained. A questionnaire that assessed psychological resilience was completed. Also, sonographic assessment of fetal biometry, cervical length and uterine artery Doppler measurements were performed.

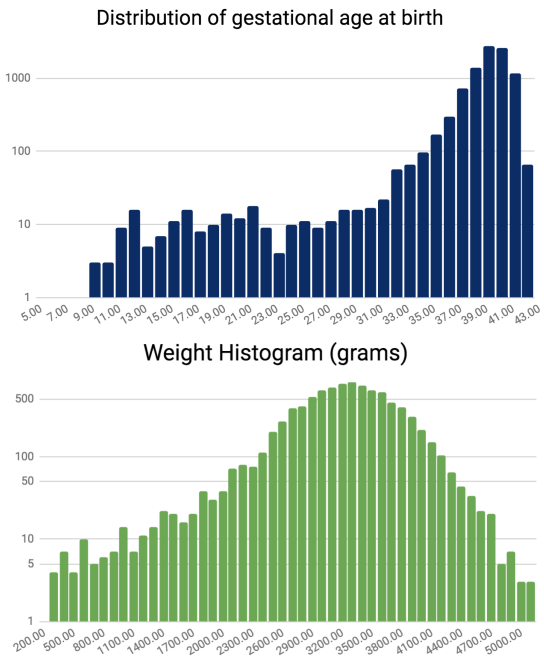


Figure 4: Gestational age and weight in the nuMoM2b data

Here are other characteristics of the nuMoM2b data: There are 10,426 features in total and 2,582 are obtained from the four visits. 1,425 of the features are qualtitative which 9,001 are quantitiative (numerical values). Missing values are codified with no information. The mean for missingness of features in the dataset is 20.68 meaning 20% of the features are missing. The median for the missingness is 1.78 which means this is a dense matrix.

4 Research Plan

4.1 Objective 1: Prediction of Preterm Birth

Our first aim is to build a series of accurate prediction models for PTB using the nuMoM2b dataset. Such models will handle the challenges common to medical datasets including (a) imbalance in the classes and missing data (Section 4.1.1), (c) correcting demographic bias Section(4.1.2), and (c) taking advantage of future data with a LUPI-based approach (Section(4.1.3).

The nuMoM2b dataset was collected at six established checkpoints during the pregnancy. We present the temporal data emissions in an HMM-like diagram in Figure 5. When a subject became pregnant at $t0$, prior information including demographic and previous pregnancy history was collected. The next four checkpoints ($t1 - t4$) were medical office visits. These visits produced a myriad of data including additional demographic data and a significant amount of bio-medical data. By the final visit ($t4$), there are a total of 2,582 feature dimensions from the four visits. The sixth time point ($t5$) was the actual delivery and included outcome results.

4.1.1 SVMs corrected for class imbalance and missingness

In the following, we will describe our proposed work to handle imbalance and missingness in the data. We frame PTB as a binary classification problem, where patients who deliver a baby preterm (full-term) are assigned the positive (negative) class respectively.

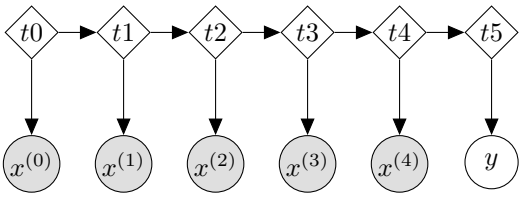


Figure 5: Temporal Data Arrangement

Preliminary Work As mentioned earlier, as part of our NSF funded EAGER grant, we conducted preliminary experiments on the MFMU data with linear and non linear Support Vector Machines (SVMs) at different times as described in Section 3, based on non-genetic maternal attributes [20, 103, 101]. Our results (Figure 6) demonstrated the superiority of non-linear methods in predicting preterm birth. Besides the fact that no time-dependent prediction was ever used on the MFMU dataset, we obtained an average of sensitivity and specificity in predicting PTB of 56% and 68% respectively, well above the performance reported in the literature on this dataset the 21% for sensitivity and 30% for specificity on average reported in the literature for this dataset [60].

	Sensitivity			**Specificity**			**g-mean**		
	T0	**T1**	**T3**	**T0**	**T1**	**T3**	**T0**	**T1**	**T3**
Preterm vs. Fullterm, All data									
Lasso	0.59	0.52	0.50	0.59	0.67	0.73	0.59 ± 0.02	0.59 ± 0.02	0.60 ± 0.03
Elastic Net	0.59	0.51	0.50	0.59	0.67	0.73	0.59 ± 0.02	0.59 ± 0.02	0.60 ± 0.03
Linear SVM	0.40	0.43	0.45	0.83	0.82	0.84	0.58 ± 0.04	0.59 ± 0.02	0.62 ± 0.03
Poly. SVM d2	0.56	0.62	0.67	0.63	0.65	0.66	0.59 ± 0.03	0.64 ± 0.01	0.6 ± 0.04
Poly. SVM d3	0.55	0.27	0.23	0.62	0.87	0.93	0.57 ± 0.03	0.49 ± 0.02	0.46 ± 0.07
RBF SVM	0.58	0.55	0.59	0.62	0.72	0.72	0.60 ± 0.04	0.63 ± 0.03	0.65 ± 0.03
Random Forest	0.3	0.3	0.3	0.86	0.88	0.93	0.51 ± 0.07	0.48 ± 0.09	0.5 ± 0.1
Creasy-7	0.30	0.22	0.21	0.88	0.91	0.93	0.52 ± 0.04	0.45 ± 0.03	0.44 ± 0.03
Creasy-13	0.29	0.31	0.31	0.86	0.89	0.91	0.50 ± 0.04	0.52 ± 0.06	0.53 ± 0.03

Figure 6: Average test rates for all algorithms at each tick of the MFMU data [101]

Background Support vector machines (SVMs) belong to the family of maximum margin classifiers. The standard approach is to solve the soft-margin formulation [24]:

$$\begin{aligned} \underset{w,b,u}{\text{minimize}} \; & \frac{1}{2}||w||_2^2 + C\sum_{i=1}^{N} u_i \\ \text{s.t.} \quad & y_i(w \cdot x_i + b) \geq 1 - u_i, \; \forall\, i \in \{1, \ldots, N\} \\ & u_i \geq 0, \; \forall i \in \{1, \ldots, N\} \end{aligned} \tag{1}$$

where C is a positive constant determining the trade-off between maximizing the margin and minimizing the mis-classifications. The u_i's are slack variables that allow to calculate the mis-classifications. An example x_i is mis-classified if its corresponding slack variable $u_i \geq 1$.

A margin error occurs if $0 \leq u_i \leq 1$. A large C corresponds to assigning a higher penalty to errors. This is useful, since in practice, data is rarely linearly separable. Typically, the SVM produces a classifier that labels examples x_i with:$f(x_i) = \text{sign}(w \cdot x_i + b)$

Limitations of current methods Support Vector Machines have shown excellent performance to model problems with linear and non-linear relationships between the predictors and outcome. They also offer theoretical guarantees, which make the method very compelling for medical applications.

However, they require learning from a full matrix of data which makes it a challenge to apply to data with missing feature values, common in most applications. The problem is accentuated when missing values occur in the support vectors and/or the mis-classified instances, leading to a biased solution. Most studies to date use imputation to deal with this problem, however, this can be time consuming and prone to errors. Furthermore, imputing data for patient, may be inaccurate. There has been a few very specific SVM missing data strategies proposed (e.g., [13, 88, 90, 67, 46]), that remain not practically usable, or too complex to be implemented and widely deployed.

Finally, class imbalance is another common problem that could bias the SVMs' results and that require some preliminary data preparation like using oversampling/under-sampling or different regularization costs.

Proposed Research Activities We provide a formulation that is able to deal systematically with missing data and class imbalance at once. Our aim is to develop an effective, efficient, and general purpose framework. We will achieve this by designing an objective function for SVMs that captures and corrects for these issues. We propose a balanced objective function for SVM handling both missing values without imputation and correcting for the imbalance in the classes as follows:

$$\begin{aligned} \underset{w,b,u}{\text{minimize}} \; & \frac{1}{2}||w||_2^2 + C\left(\frac{\sum_{i=1}^N \mathbf{1}[y_i=+1]u_i}{\sum_{i=1}^N \mathbf{1}[y_i=+1]} + \frac{\sum_{i=1}^N \mathbf{1}[y_i=-1]u_i}{\sum_{i=1}^N \mathbf{1}[y_i=-1]}\right) \\ \text{s.t.} \quad & y^{(i)}\left(b + \dim(x)\frac{\sum_{j=1}^{\dim(x)} w_j \cdot x_j^{(i)} \cdot z_j^{(i)}}{\sum_{j=1}^{\dim(x)} z_j^{(i)}}\right) \geq 1 - u_i, \; \forall\, i \in \{1,\ldots,N\} \\ & u_i \geq 0, \; \forall i \in \{1,\ldots,N\} \end{aligned} \tag{2}$$

where $z^{(i)} \in \{0,1\}^{\dim(x)} \; \forall\, i \in \{1,\ldots,N\}$, and where a 0 indicates missing data for $x_j^{(i)}$, and vice-versa for $z_j^{(i)} = 1$. The total number of features is $\dim(x)$.

The objective function shown above has two main differences from the vanilla SVM. First, it tries to resolve data imbalance by putting less weight on the majority class $(y = -1)$ and more weight on the minority class $(y = +1)$. Observe in the objective function how the sum of the slack values for each class are each being divided by the number of examples for each class. This change allows the minority examples to have more effect on the slack term in the objective function than the majority examples.

Another difference is how the function handles missing data. In the first constraint, we introduce a new matrix z: $\dim(x)\frac{\sum_{j=1}^{\dim(x)} w_j \cdot x_j^{(i)} \cdot z_j^{(i)}}{\sum_{j=1}^{\dim(x)} z_j^{(i)}}$. z is a 2D matrix with same dimension as x. This z matrix essentially allows us to ignore missing values when calculating $w \cdot x_i$ term in the constraint formula. No matter what the missing values actually are, it is being multiplied by zero, and hence the missing values do not matter anymore. It compensates the ignored missing values (by multiplying by zero) by putting more weight on the available data. This is done by multiplying $w \cdot x_i$ by (total num of features)/(number of features available) which corresponds to $\frac{\dim(x)}{\sum_{j=1}^{\dim(x)} z_j^{(i)}}$.

We will implement this strategy and compare to the state of the art, including the complete case as a baseline, mean imputation, multiple imputation, K-nearest Neighbor imputation, etc. We will analyze our method under different settings of missing data. We will finally conduct experiments on the nuMoM2b dataset with and without imputation.

4.1.2 Correcting Demographic Bias and data disparity

Just as we normalized the slack penalty with respect to the sample size of each outcome class, so too we can construct an SVM objective that normalizes its loss with respect to the sample size of protected

demographic classes [57]. First, we define an objective function that operates over a sequence of samples and their associated slack terms $S = \{(x_i, y_i, u_i)\}_{i=1}^{|S|}$:

$$f(S) = \left(\frac{\sum_{i=1}^{|S|} \mathbf{1}[y_i = -1]u_i}{\sum_{i=1}^{|S|} \mathbf{1}[y_i = +1]} + \frac{\sum_{i=1}^{|S|} \mathbf{1}[y_i = -1]u_i}{\sum_{i=1}^{|S|} \mathbf{1}[y_i = -1]} \right)$$

We can then partition the minimization objective to operate over subsets of S that are conditioned on demographic criteria being satisfied (e.g. white vs. non-white).

$$\begin{aligned} \underset{w,b,u}{\text{minimize}} \; & \frac{1}{2}||w||_2^2 + C\Big(f(\{S : \text{demographic condition}\}) + f(\{S : \neg\text{demographic condition}\})\Big) \\ \text{s.t.} \quad & y_i(w \cdot x_i + b) \geq 1 - u_i, \; \forall\, i \in \{1, \ldots, N\} \\ & u_i \geq 0, \; \forall i \in \{1, \ldots, N\} \end{aligned} \tag{3}$$

This partitioning effectively normalizes the demographic subsets' contribution to the minimization objective. Given the demographic biases present in our data, we want a formal way to ensure unbiased models are created. We will implement and extensively test this approach on different demographic condition, such as the race.

4.1.3 Learning Using Privileged Information

Given the temporally evolving nature of the data, we believe we can bolster the performance of these temporal model instantiations by training on both observable (past and present) data and unobservable (future) data while only predicting on the observable. In this manner, the unobservable future data can be considered privileged and thus we can apply the Learning Using Privileged Information (LUPI) paradigm [66]. This framework leverages the use of privileged information, and shows significant promise in achieving higher performance. SVM+ is a known method for implementing the LUPI paradigm [99], which can be formulated as a convex optimization problem (equation 4). Here, x^* is privileged information used only during training. Note that this formulation assumes $y_i \in \{-1, 1\}$.

$$\begin{aligned} \underset{w,b,w^*,d}{\text{minimize}} \; & \frac{1}{2}||w||_2^2 + \frac{\gamma}{2}||w^*||_2^2 + C\sum_{i=1}^{N}(w^* \cdot x_i^* + d) \\ \text{s.t.} \quad & y_i(w \cdot x_i + b) \geq 1 - (w^* \cdot x_i^* + d), \; \forall\, i \in \{1, \ldots, N\} \\ & w^* \cdot x_i + d \geq 0, \; \forall i \in \{1, \ldots, N\} \end{aligned} \tag{4}$$

Predictions are made using only non-privileged information, which is also the prediction formula for regular SVM [25]. Comparing SVM+ (equation 4) with regular SVM (equation 1) is useful in understanding how privileged (future) data may produce a better classifier. Again, our SVM formulation assumes $y_i \in \{-1, 1\}$. Observe how SVM+ replaces the slack variables, u_i, with an affine operation on the privileged features. In essence, future data will govern the slack applied to each training example such that a better decision boundary might be learned. Outside of this intuition, SVM+ is known to have better convergence properties [66]. Specifically, if the error-correcting space (privileged data) is good, then the convergence (w.r.t. training set size) in the combined space could be on the order of $1/n$, as opposed to an algorithm operating only in the decision space (observable data), which converges on the order of $1/\sqrt{n}$, amounting to a difference of needing **320 training examples versus 100,000** training examples, which is a tremendous difference. This is a very compelling property in our application given the limited number of training examples. In fact, this is compelling for many other medical applications with a scarce number of patients to train prediction models!

Proposed Research Activities This LUPI property offers strong theoretical support for our application of SVM+ to construct a robust classifier that leverages future data at training time. The original work [102] showed for SVM-based models that prediction performance generally improves as more features become available. This improvement implies that the additional features are important for making better predictions. Therefore, we hypothesize that future data can be used as an error-correcting space to improve the convergence properties of the classifier.

A model trained at time, t, will declare all data collected at or before t as the decision space (observable data upon which predictions are made). Similarly, a model built at time t, will utilize future data as the error correcting space (unobservable data used only for training–not for prediction). This process is defined formally in set notation in equation 5. Note that we drop the (t) in all other formulas for notation simplicity. Also, when all data becomes observable, we replace the privileged feature space with a constant value of 1 for comparative purposes. We note however, that this replacement will inhibit SVM+ to optimize per-example slack variables. In this case, SVM+ can optimize global slack via w^* and d. This insufficiency is inconsequential since one would elect for regular SVM when there is no access to an error correcting space.

$$\begin{aligned} x(t) &= \{x^{(j)} \mid j \leq t\} \\ x^*(t) &= \begin{cases} \{x^{(j)} \mid j > t\} & , \ t \leq t4 \\ 1 & , \text{o.w.} \end{cases} \end{aligned} \tag{5}$$

We will enhance SVM+ with label balancing and also correct for the missingness.

$$\begin{aligned} \underset{w,b,w^*,d}{\text{minimize}} \ & \frac{\gamma_w \cdot ||w||_2^2}{\dim(x)} + \frac{\gamma_{w^*} \cdot ||w^*||_2^2}{\dim(x)} + \frac{\sum_{i=1}^N \mathbf{1}[y_i = +1]u_i}{\sum_{i=1}^N \mathbf{1}[y_i = +1]} + \frac{\sum_{i=1}^N \mathbf{1}[y_i = -1]u_i}{\sum_{i=1}^N \mathbf{1}[y_i = -1]} \\ \text{s.t.} \quad & y_i(w \cdot x_i + b) \geq 1 - u_i, \ \forall \, i \in \{1, \ldots, N\} \\ & u_i \geq 0, \ \forall i \in \{1, \ldots, N\} \\ & u_i = w^* \cdot x_i^* + d \end{aligned} \tag{6}$$

In addition to SVM+ (equation 6), we also consider as a baseline, Logistic Regression with l-2 regularization (equation 7 where $y_i \in \{0, 1\}$) and regular SVM (equation 1). We consider these alternative convex models both to measure any improvement with SVM+ and to confirm the results of the original work [102].

$$\underset{w,b}{\text{minimize}} \ \gamma ||w||_2^2 + \sum_{i=1}^N \ln(1 + \exp(w \cdot x_i + b)) - y_i(w \cdot x_i + b) \tag{7}$$

For SVM and SVM+, we also consider Gaussian kernels. At training time, we replace x_i with $K_{:,i}$, the corresponding column of the Kernel matrix ($K \in \mathbf{S}_+^N$) where each entry is computed according to equation 8.

$$K_{i,j} = \phi(x_i, x_j) = \exp\left(- \frac{||x_i - x_j||_2^2}{2} \right) \tag{8}$$

To make a prediction for a new sample, x', we construct a "kernelized" feature vector according to equation 9, where $x_1 \ldots x_N$ correspond to training examples (i.e. those that may form support vectors).

$$K' = \begin{bmatrix} \phi(x', x_1) \\ \vdots \\ \phi(x', x_N) \end{bmatrix} \in \mathbf{R}^N \tag{9}$$

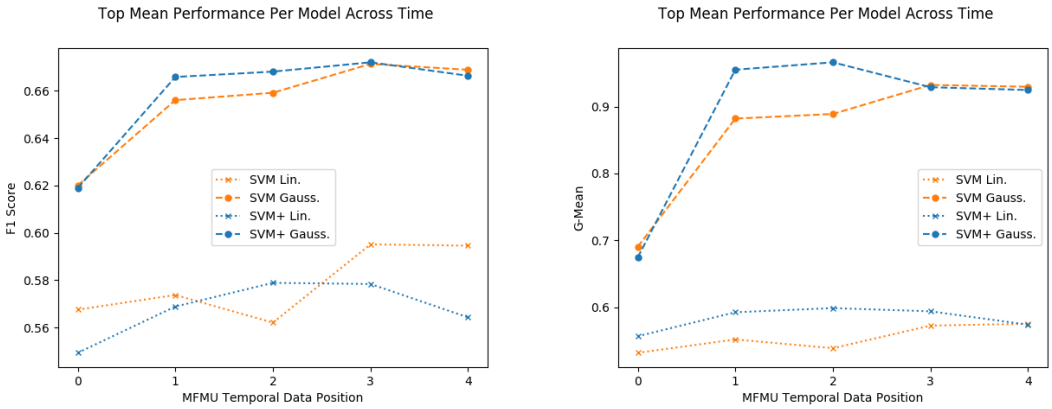


Figure 7: Mean F1 and the geometric mean of sensitivity and specificity cross validation results for the best hyperparameters over time for the MFMU.

Preliminary Work We used the MFMU data to conduct a preliminary study on the usefulness of SVM+ for PTB.

Our original work [102] only trained models for $t0$, $t1$, and $t3$ using SVM with different kernels, logistic regression, random forest, and Creasy table as a baseline. We report our new unpublished results on using SVM+ using the MFMU in Figure 7. These represent cross-validation results.

As hypothesized, SVM+ generally outperforms the other models. In fact, SVM+ with the Gaussian kernel either comes in first or ties for first (with respect to F1 and GMean) at all time steps. SVM+ with a linear kernel also does well, however, we know from [102], that there is no good solution in the linear space to begin with. LUPI is good to use, when there exist a good solution in the x space [98].

Thus, it seems that the privileged future data constitutes a good error correcting space for the PTB prediction problem.

We also examine the mean F1 performance trajectory across time for the five model-kernel combinations. First, notice that all models without access to the privileged (future) error correcting data space naturally improve with time as the data becomes richer. Conversely, SVM+ with a Gaussian Kernel is at peak performance throughout, seeing almost negligible performance gain through $t3$ and a negligible drop at $t4$.

4.2 Objective 2: Evaluating the Contribution of Genetics to PTB

4.2.1 Overview: Combining Standard Paradigms with Methodological Innovation

This aim leans on newly collected data that enables a genomewide association study (GWAS) of PTB, to investigate the genetic component of various aspects of this phenotype. Our analysis approach to this data is two-phased. First, we will apply standard GWAS analysis, in order to provide timely feedback to collaborators as well as serve as baseline for additional development. Next, we will seek to develop improved analysis of involvement of genetic contributions in PTB. This builds on specific and unique characteristics of this trait and phenotypically rich dataset. Specifically, the temporal nature of accumulated data throughout pregnancy facilitates tackling questions of causality between the genetic information and its various forms of phenotypic implications.

4.2.2 GWAS for PTB on the nuMoM2b Cohort

The nuMoM2b cohort was recently genotyped for $2.5M$ Single Nucleotide Polymorphism (SNP) markers throughout the genome. Data production an completed by University of Indiana and is to be shared with the

project team promptly, prior to public release. The data will undergo stringent QC, followed by ancestry assignment based on Principal Component analysis (PCA) guided by 1000 genomes data. Ancestry labels will be assigned by cluster analysis. Based on self reported demographic data in this cohort, we expect statistically powered ancestry cohorts for each of European, African American, and Hispanic ancestries. Samples with other labels or with labels discordant with self reported demography will be excluded. Imputation of all variants of frequency $> 0.5\%$ will be performed using IMPUTE2 [45] without pre-phasing, a strategy proven effective in diverse cohorts [79]. We use PTB as a dichotomous phenotype as the primary phenotype definition, using an additive genetic model with covariates as the features available at visit 1. Covariates would be linear, with the standard exception of age^2 as an additional covariate. Secondary analyses will explore extreme PTB as a dichotomous phenotype, gestational length as a quantitative phenotype, and test each with covariates available after each visit, and with non-additive genetic models. Association of each variant will be performed in parallel for each of the three ancestry groups, adjusted for inflation due to genetic structure using a linear mixed model by the EMMAX package [49]. We will combine z-score association results across all ancestry groups by meta-analysis using METAL [106]. We will compute β association coefficients to each SNP, identifying genomewide significant loci. Bases on an existing GWAS for PTB [110], we expect the EBF1 locus, but likely no other, to be genomewide significant in our European-ancestry cohort. We will specifically attempt to replicate known associations to PTB and related phenotypes, and to replicate our own findings in dbGaP studies. Importantly, we will derive a Polygenic Risk Score (PRS) that is corrected for linkage disequilibrum using the LDSC package [12].

Even this vanilla application of standard GWAS analysis pipelines will suffice to provide new insight regarding the changing patterns of genetic association as additional phenotypic data is accumulated. This analysis would thus serve as a baseline for evaluation of the utility in more advanced developments.

4.2.3 Incorporating Genetic Features into LUPI

LUPI methods are designed for moderately high dimensionality of the feature space. The millions of SNP markers can thus not be all used as features. We will consider only the genotypes of each SNP associated with PTB, whether previously known, or newly discovered, as a feature, known prior to visit 1. The genotypes will be linearly standardized to mean zero and variance one: this does not matter for the linear kernel, but makes distances between feature vectors more comparable, thus should help the Gaussian kernel. We will further consider the PRS for PTB as a feature, accounting for the genetic contribution from the rest of the genome. We will revise LUPI analysis, adding these extracted features as predictors.

This direct application of our developed methods enables direct interrogation of the changes in importance of genetic features as parts of the SVM discriminator. We will evaluate the temporal improvements in prediction accuracy across different visits, and vs. the baseline predictions.

4.2.4 Mendelian Randomization for Classifier Features

We will directly evaluate the extent genetic information has value in predicting PTB even after visit test results are available. Conditional independence of PTB on a genetic feature given visit data would mean that such feature is valuable only pre-visit, motivating the acquisition of the better discriminant features. It would also implicate these features as parts of the mechanism by which this feature is contributing to PTB. In contrast, the dependence on a feature even when conditioning on visit data implies it needs to be further considered even later during pregnancy, and implicate its effect pathway as mediating visit results and PTB.

We will formally evaluate the conditional contribution of genetic information by adding Platt scaling to the SVM optimization target, allowing its interpretation as a probability of class assignment. This facilitates quantification of the importance of each genetic feature in terms of the probability change due to its contribution.

4.3 Objective 3 : Clinical and Social Impact

While the goal of objective 1 is to both improve the prediction models we build, in this objective we test the effectiveness of the models and use what we learn from the models to improve patient care as it relates to visits and diagnostics tests etc. This will help us to identify women who are at the highest risk for extremely preterm birth in order to facilitate interventions as well as identify women at the lowest risk to avoid unnecessary and costly interventions. We plan to assess the effectiveness of the methods in clinical practice by testing the effectiveness of the longitudinal models produced in Objective 1 and 2 on existing clinical data. We (b) building a sequential decision making model; this includes optimizing the scheduling of patient visits as well as diagnostic testing. We will study whether these activities can be tailored for different classes of patients.

We will test our models on auxiliary datasets that are available through Dr. Wapner's lab. In addition, we will closely work with our New York Presbyterian Hospital collaborators (see Letter attached) to implement and test the longitudinal models devised in this project in their EHR systems.

Previous research [87, 33, 7, 18, 89, 6] has recognized and studied the role as well as challenges of capturing simple and complex trends for medical decision-making. Scheduling doctors visits for pregnant women either in a predetermined inflexible way or in a purely reactive fashion could lead to undesirable results, depending on the sequence of events in the patient's history, or the sub-population. This is also true for scheduling blood and other tests for pregnant women during the course of their pregnancy. An effective PTB visit scheduling and diagnostic testing plan requires dynamic planning and reasoning in the context of uncertainty.

4.3.1 Background

Sequential decision-making is required when there are a large number of embedded decision nodes in the branches of the decision tree. Optimizing the frequency of doctor's visits for pregnant women as well as the diagnostic testing regime based on a patient's risk of PTB can be viewed as an instance of sequential decision-making. Markov decision processes (MDPs) are powerful analytical tools used for sequential decision making and optimal planning in stochastic environments. They have been widely used in many industrial and manufacturing applications. An MDP is a probabilistic model of a sequential decision problem, where states can be perceived exactly, and the current state and action selected determine a probability distribution on future states [70, 92]. Specifically, the outcome of applying an action to a state depends only on the current action and state (and not on preceding actions or states). Co-PI Raja has extensive experience using MDPs and DEC-MDPs (Decentralized Markov Decision Processes) for decision support in both single agent as well as distributed multiagent environments [16, 17, 27, 71, 72].

4.3.2 Proposed Research Activities

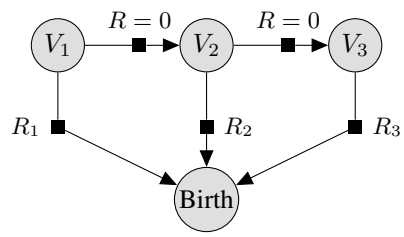
We propose a decision-theoretic approach for (sequential) temporal action policies that account for both dynamic temporal trends in patient history, as well as more static social-cultural descriptors. nuMoM2b provides a temporal view of a patient's evolving state of health, through symptoms, lab results, medical actions and results. This investigation will identify latch points that form the basis for preliminary design and proposals for visit scheduling and diagnostic testing policies. Developing scheduling policies for visits and tests based on individual patient needs would involve among other things determining the correct level of abstraction for state features, accurately identifying actions and their interdependencies, determining the non-stationary transition probability function as well as the appropriate intermediate and goal state rewards. We will build on our preliminary work to define the following finite horizon MDP model. In most medical problems, transition probabilities are non-stationary [6, 31] such as when the transition probability represents the probability of a diagnosis that is increasing over time or when age dependent diagnosis is involved.

This is the case in the PTB problem since age plays a role in PTB diagnosis and visit schedule. We will study the applicability of existing methods like natural history computation [6] and Weibull distribution [105] in the PTB domain to derive probabilities from the datasets. $R(s_i, a, s_j)$ is the reward obtained in state s_j as a consequence taking a prevention action a in state s_i. The MDP will have both intermediate and goal state rewards since intermediate actions will have observable results in the near-term while the goal state reward is determined on whether the birth is full-term or not. The reward function will have to account for both the benefits (health and societal) of an intervention as well as the costs. This is also called cost-effectiveness analysis. We will then compute the optimal stochastic prevention policy for the MDP model.

Figure 8 shows an example Markov Decision Process (MDP) formulation for PTB, in which a mother attends 3 office visits. At each visit and based on the observations available, the agent will decide between two actions: (1) Do not begin interventional treatment and proceed to the next visit; (2) Begin interventional treatment and proceed to birth outcome. This MDP makes the critical assumption that interventional treatment once begun cannot be halted. Circumventing future visits (i.e. $V_1 \rightarrow$ Birth) is a modeling convenience and will not reflect reality as a mother will obviously still likely to continue visiting her physician throughout her pregnancy.

There are numerous intuitive ways to devise an appropriate reward structure. Let's first consider the economics of PTB. We define $\mathbb{E}[\text{cost of care}]$ to be the expected post-birth costs to treat and care for a prematurely born infant while $\mathbb{E}[\text{cost of intervention} \mid \text{time}]$ is the expected cost of beginning pre-birth interventional treatment at a particular point during pregnancy.

Figure 8: Visit Schedule MDP



We can use these to formulate a possible reward structure:

$$R_t = \begin{cases} +\mathbb{E}[\text{cost of care}] & \text{, if true positive} \\ -\mathbb{E}[\text{cost of care}] & \text{, if false negative} \\ +\mathbb{E}[\text{cost of intervention} \mid \text{time} = t] & \text{, if true negative} \\ -\mathbb{E}[\text{cost of intervention} \mid \text{time} = t] & \text{, if false positive} \end{cases} \tag{10}$$

These are just a starting example for a visit scheduling policy. There is the obvious social question of whether optimizing the economics of this malady is acceptable. Normalizing economic costs and survival rates would allow us to integrate the two reward structures and balance them as deemed appropriate. Models for diagnostic testing options can be augmented into this model using similar reasoning.

We plan to validate the policies that result from this work with both medical and social experts scoring it. We also plan to collect evidence that documents the effectiveness of current policies with regard to visit schedules and testing/treatment regiments. Also in the health field, ethical and legal regulations make it necessary to do sensitivity analysis to ensure that the uncertainty in the decision model does not adversely affect the conclusions of the analysis [19]. A potential challenge is whether the nuMoM2b data will allow us to build a sufficiently rich MDP model (accurate transition probabilities and meaningful reward functions) to produce prevention policies that medical decision makers will find effective. We will harness results from existing literature to understand the accuracy and richness of our model.

5 Team, prior NSF support and timeline

The multidisciplinary composition of the research team and their existing collaborative relationship uniquely qualify them for the proposed work. The team has already acquired the data and conducted some exploratory

data analysis and meta data. Codebooks were created to document all pieces.

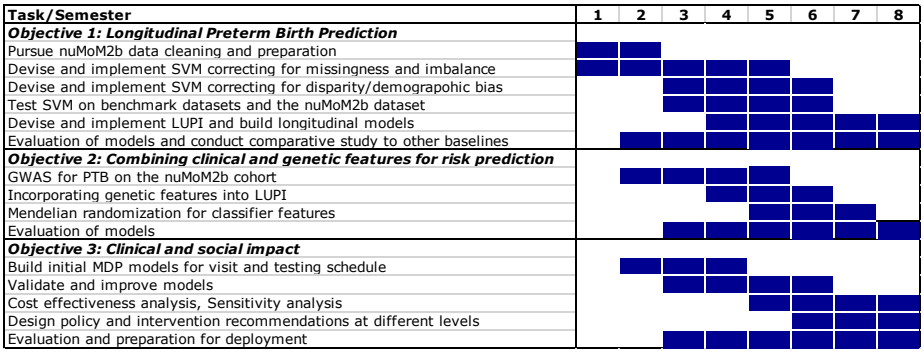


Figure 9: Timeline of the project Milestones

The PI is a Lecturer in Discipline with expertise in machine learning methodologies and applications. Her work includes optimization of quantitative features in association rules[86, 85], frequent pattern mining[82, 97], action recommendation[96], sequential event prediction[80], maximum entropy density estimation [46], crowdlabeling [54, 52] and clinical informatics [2, 84]. She worked on several machine learning applications [83, 97, 95, 47, 40, 81]. PI Salleb-Aouissi has previous funding from NSF: (1) *"EAGER: Collaborative Research: Advanced Machine Learning for Prediction of Preterm Birth"* (Award # 1454855), Sept. 2014 - Aug. 2017, $175,457 This lead to one female Ph.D. student defense, and several publications. We did exploratory work to develop a framework for data collection through labeling of unstructured data [51, 55, 56, 53] and design a preliminary risk scoring system for PTB prediction using the MFMU dataset [101, 103]. Our initial results were promising and has led to the development of this proposal.

Co-PI Pe'er is an expert in computational methods to study human genetics. Co-PI Pe'er previous NSF funding: *EAGER: Topic Models for Population Genetics* (Award #: CCF-1547120) Amount: $200,000, Jul 2015- Jun. 2018. Pe'er developed Mixed-Layer Analysis of Genetics and Expression (MILAGE), a model that combines direct effects of cis-SNPs on nearby transcripts with trans-effects that control global factors of expression in a tissue-specific pattern.

Co-PI Raja is an expert in distributed intelligent systems and meta-level reasoning. Her work involves leveraging decision theoretic methods to model decentralized control [74, 104] and leveraging state abstractions and offline multiagent reinforcement learning to optimize reward functions of any shape (monotonic or otherwise) for metareasoning [73, 8, 9, 75, 27]. Co-PI Raja's previous NSF funding: (1) *"EAGER: Collaborative Research: Advanced Machine Learning for Prediction of Preterm Birth"* (Award # 1454814), Sept. 2014 - Aug. 2017, $30,643. Described above.
(2) *"RI: Small: A Hybrid Approach For Meta-Level Control Across Agent Boundaries"* (Award # 1018067), Sept. 2010 - Dec. 2013, $167,653 that included an REU supplement for 2 undergraduates. Raja led the design of novel meta-level control algorithms for multiagent tornado tracking using decentralized learning and conflict resulting in multiple publications [14, 15, 17, 16, 75, 76]. This project supported one Ph.D. students and two REU students who were participants in the NSF-funded STARS program at UNC Charlotte.
(3) *"Doctoral Mentoring Consortium at the Twelfth International Conference on Autonomous Agents and Multi-Agent Systems"* (Award # 1308505), May 2013-April 2014, $20,000. This award supported the travel and participation costs of a diverse group (20) US-based young researchers, including women and members of underrepresented groups, in the AAMAS 2013 and the doctoral consortium held in conjunction with it.

Co-PI Wapner is a very successful obstetrician and principal investigator of several NIH projects. He has a keen interest in machine learning methods to understanding etiologies of adverse pregnancy outcomes.

Co-PI Friedman is an Assistant Professor of Obstetrics & Gynecology at Columbia University Medical Center. His research interests include cost-effective care quality in obstetrics, prenatal diagnosis, and health care screening.

Data Management Plan

Proper storage and archiving of data is important to this research. The nuMoM2b dataset was provided to Dr. Wapner from RTI International on 2/6/2018. Any use of this data must be in accordance with the Data Transfer Agreement (DTA) between RTI and Columbia University. This data management is designed to satisfy the above-mentioned DTA, NSF's data sharing policy, and the open access policies of our institutions.

Types of Data The nuMoM2b dataset was collected across 8 main sites, including Columbia University (PI: Wapner), where 10,038 nulliparous women were enrolled between Sept 30, 2010 and September 23, 2013, and followed prospectively from early during the pregnancy to delivery. At this point of time, the data was released only to the clinical study investigators of the nuMoM2b study, to provide the opportunity for local analysis of the data, and increase the scientific output of the study. The data produced as part of this project include: software and documentation, curricular materials and archival publications. The software generated in the form of source and compiled code in various programming languages (including Java and Python)

Access. During the course of the research project, the PIs will be responsible for adhering to data retention and security standards. The research products, whereever appropriate, will be shared with the general public through the website. The PIs and student researchers will together determine the software components shared under this project. The public access will be regulated by the universities in order to protect privacy and confidentiality concerns and to respect any proprietary or intellectual property rights. Our project is under research protocol number IRB-AAAR9413 approved by the Columbia University Institutional Review Board (IRB) on October 2018, 2018 and under research protocol number IRB-55-2018 approved by the Cooper Union on November 7, 2018. Data currently resides on encrypted end points. Graduate research assistants will have the appropriate HIPAA training to access the data. Columbia University IRB will be overseeing all aspects related to data security. Coordinator DiVito from Dr. Wapner's office, will be overseeing all aspects related to data sharing and obtain the appropriate Data Transfer Agreement from RTI International. DiVito will also obtain agreements required for the genetic data transfer between the University of Indiana, and Columbia University along with obtaining permission for publishing our results. The team is already using Gitlab to share all code and data code books. We expect the code developed in the context of this project to implement (1) An SVM robust to missing data, class imbalance, and bias, (2) LUPI-based SVM including (3) a sequential action recommendation approach. Our doe implementations will be available as open source software and shared via Github.

Dissemination. The results from this research will be published in international AI and ML journals and conferences (AAAI, ICML, MLHC, IJCAI) and in workshops associated with those conferences. The data produced as part of this project will be made publicly in our existing website for the project (`http://www.cs.columbia.edu/~ansaf/cing/PTB/`). Results will further be disseminated through progress reports to NSF and news postings on the PI research websites.

Policies for Reuse and Redistribution Software will be available, in source code, under an open source software distribution license, such as the GNU General Public License (that allows free use and modification of software as long as the result is released under the same license) which will allow other researches to validate and build on this project's results. Terms of use will include requirements of attribution along with disclaimers of liability. The PIs will retain principal legal rights to the intellectual property developed under this grant as is compliant with the home institution.

Retention All code will be retained and made available on the project website for a minimum of three years after the conclusion of the project. The project website and data will be managed on the PI Salleb-Aouissi's laboratory servers and in compliance with data transfer agreements. Research data that supports

patents will be retained fro the entire term of the patent. Final peer-reviewed manuscripts and supplemental information such as data tables will be posted to the NSF public access repository as well. They will be available within a year of publication.

Collaboration Plan

This project is a collaboration among four researchers. Together they provide expertise in machine learning, intelligent decision support, obstetrics and preterm birth, electronic health records, and genetics. The PIs will work closely to address this challenging problem of predicting preterm birth.

PI Salleb-Aouissi is a Lecturer in Discipline in the Department of Computer Science at Columbia University. Her expertise lies in Artificial Intelligence and Machine Learning and her research is on AI and education and medical informatics.

PI Raja is a Professor of Computer Science at the Cooper Union. She has developed decision support algorithms context of uncertainty and limited data and computational resources. Her expertise is in sequential decision making and bounded rationality.

PI Wapner is Professor of Obstetrics and Gynecology, Maternal-Fetal Medicine and Director of the Division of Translational and Clinical Research at Columbia University. He is also PI for the Columbia center of the NIH-sponsored 'Preterm Birth in Nulliparous Women' (nuMoM2b) study. Dr. Wapner was one of the pioneers of chorionic villus sampling and first trimester Down Syndrome screening and has been a Principal Investigator (PI) in the MFMU Network for almost 15 years. He has led multiple national multicenter clinical studies that have resulted in significant changes in clinical care.

PI Alexander Friedman is an Assistant Professor of Obstetrics & Gynecology at Columbia University Medical Center. His research interests include cost-effective care quality in obstetrics, prenatal diagnosis, and health care screening.

PI Pe'er recently joined the project and will contribute to objective 2 to include genetic factors to the risk prediction models. Professor Pe'er is an associate professor in the Department of Computer Science at Columbia University. His laboratory develops and applies computational methods for the analysis of high-throughput data in germline human genetics.

PIs Salleb-Aouissi, Raja, Wapner, Friedman, and Pe'er will collaborate on all proposed objectives. We expect this collaboration to succeed since several of the PIs already have a history of working together. Salleb-Aouissi and Raja have met bi-weekly since the start of their collaboration on this work in early 2012 and Raja was a visiting faculty at Columbia University's CCLS lab from 2011 to 2012. Along with PI Wapner, they were PIs of a NSF Eager grant on prior work on this topic and together they have co-authored papers [20, 103, 101, 100].

PI Pe'er has attended several of the meetings and have brainstormed extensively in the development of this proposal. PI Salleb-Aouissi was the Ph.D. advisor for a female student on this project. PI Raja served as an external member on the dissertation committee of this Ph. D. student. The PIs have co-advised Masters and undergraduate students on this project as well and currently have 1 graduate student from Columbia University, and an undergraduate student each from Columbia university and Cooper Union.

All five PIs expect to work closely on all 3 objectives to close the loops on algorithm development and evaluation on the nuMoM2b dataset. To facilitate collaboration, we will continue use of shared database, code base (using our joint code repository). We will continue to have our bi-weekly in-person meetings of all PIs and/or teleconferences for progress updates, data discussions, algorithm specification and development, code reviews and analysis, development of research skills for doctoral students such as reading and presentation of related work, working on conference publications and presentations and providing research experiences for undergraduate students. PIs Salleb-Aouissi and Pe'er's offices are located in Columbia University while PI Wapner's office is in the Columbia University medical center. PI Raja is at Cooper Union in New York City. Project meetings will generally be held in the Computer Science Department at CU or the medical center. These campuses are in New York City and connected by public transportation and shuttles.

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